

Notes on lecture 1

Jiten Chandra Kalita and Shyamashree Upadhyay

We are going to do “Multivariable Calculus” in this part of the course MA 101. In this lecture, we will start with “vector calculus”.

The word “Multivariable” means more than one variables. In this course, we are going to deal with functions of more than one variables. The very first topic in the syllabus is “vector calculus”. Vector calculus is going to introduce us to geometry in 2 and 3 dimensions. There are many geometric sets that we are going to deal with in this course.

1 Review of vectors

Question: What is a vector?

Answer: A vector is a physical quantity which has both magnitude and direction.

Example 1.0.1. Suppose we consider the motion of a particle, which is moving from one point A to another point B . This is designated by an arrow joining A and B (with the arrow head pointing towards the point B). This arrow is called the **displacement vector**. The point A (from where the particle started with) is called the **initial point** of the vector. And the point B is called the **terminal point** of the vector. The distance covered from A to B is called the **magnitude of the vector**.

Generally, this vector is denoted by $\overrightarrow{AB}$ and its magnitude is denoted by $|\overrightarrow{AB}|$. Once the particle moves from the point A to B , that is going to give us the direction also. $\square$

In this course, we will indicate a vector by putting an arrow over some mathematical symbol.

Definition 1.0.2. Let us consider two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ such that their lengths (or magnitudes) are the same and they are parallel. That means, the directions of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are the same and their magnitudes are also the same.

Then we say that the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are **equivalent** and denote it by $\overrightarrow{AB} = \overrightarrow{CD}$.

Note: The initial and terminal points of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ need not be the same!

Figure 1.0.1 below illustrates the above definition of equivalent vectors.

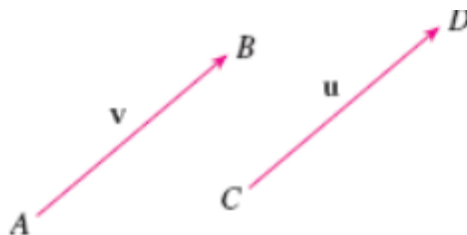


Figure 1.0.1: Equivalent vectors

1.1 Combining vectors

Definition 1.1.1. Consider 2 vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$ such that the terminal point of $\overrightarrow{AB}$ is the same as the initial point of $\overrightarrow{BC}$. If we now join the initial point A of $\overrightarrow{AB}$ and the terminal point C of $\overrightarrow{BC}$, then we get a new vector $\overrightarrow{AC}$, which is called the **sum** of the two vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$. We write

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}.$$

Figure 1.1.1 below illustrates the above concept of the sum of two vectors. The above definition gives one way (namely, **addition**) of combining 2 vectors. Now, a natural question arises.

Question: In the above definition, we had taken two vectors $\overrightarrow{AB}$ and $\overrightarrow{BC}$ such that the initial point of one vector equals the terminal point of the other vector. What happens when the two vectors that we consider donot have this property?

Answer: Let us take two vectors $\vec{a}$ and $\vec{b}$ which do not have this property. But we have learnt the concept of equivalent vectors in Definition 1.0.2 above.

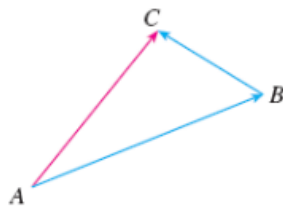


Figure 1.1.1: Sum of two vectors

Therefore, we can do some kind of a “parallel displacement” of one of the vectors (say, $\vec{b}$) and bring it to a position in such a way that the terminal point of $\vec{a}$ equals the initial point of $\vec{b}$. Then we can take their sum and call it $\vec{a} + \vec{b}$.

Definition 1.1.2. If $\vec{b}$ is a given vector, then $-\vec{b}$ is going to be the vector having the same magnitude as $\vec{b}$ but having opposite direction.

If we consider another vector $\vec{a}$ and join the initial point of $\vec{a}$ with the terminal point of $-\vec{b}$, we get a new vector which is denoted by $\vec{a} - \vec{b}$. This new vector so obtained is called the **difference** or the **subtraction** of the two vectors $\vec{a}$ and $\vec{b}$.

Figure 1.1.2 below illustrates the above concept of the subtraction of two vectors.

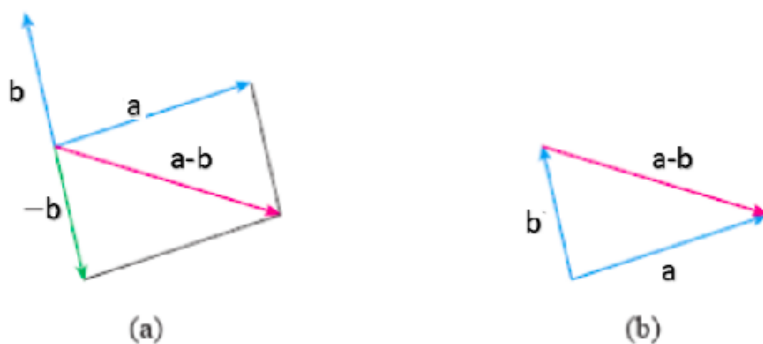


Figure 1.1.2: Subtraction of two vectors

Definition 1.1.3. Consider a vector $\vec{a}$. We can multiply the vector $\vec{a}$ by a nonzero scalar λ . Then $\lambda\vec{a}$ is a new vector whose magnitude is equal to $|\lambda|$

times the magnitude of $\vec{a}$. And the direction of the new vector $\vec{\lambda a}$ equals that of $\vec{a}$ if $\lambda > 0$ and opposite to that of $\vec{a}$ if $\lambda < 0$. This is called **scalar multiplication** of a vector $\vec{a}$ by a nonzero scalar λ .

Figure 1.1.3 below illustrates the above concept of multiplication of a vector by a scalar.

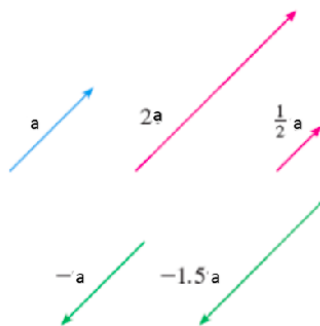


Figure 1.1.3: Scalar multiplication

1.2 The triangle law

Consider 2 vectors $\vec{AB}$ and $\vec{BC}$ such that the initial point of $\vec{BC}$ equals the terminal point of $\vec{AB}$. Then the sum of the two vectors $\vec{AB}$ and $\vec{BC}$ is equal to the third side of the triangle formed by the vectors $\vec{AB}$, $\vec{BC}$ and $\vec{AC}$. Figure 1.2.1 below illustrates the above concept of the triangle law.

1.3 The parallelogram law

Suppose $\vec{a}$ and $\vec{b}$ are two arbitrary vectors in a plane. By doing a parallel displacement, we can bring the vector $\vec{b}$ to a place such that the initial points of $\vec{a}$ and $\vec{b}$ are the same. Then we can construct a parallelogram whose sides are given by $\vec{a}$ and $\vec{b}$. Then the diagonal of this parallelogram is going to designate the sum of the two vectors $\vec{a}$ and $\vec{b}$.

Note: The triangle law and the parallelogram law can be realized for subtraction ($\vec{a} - \vec{b}$) of two vectors as well. The only point is that, one has to take a vector in the direction opposite to that of $\vec{b}$.

Figure 1.3.1 below illustrates the above concept of the parallelogram law.

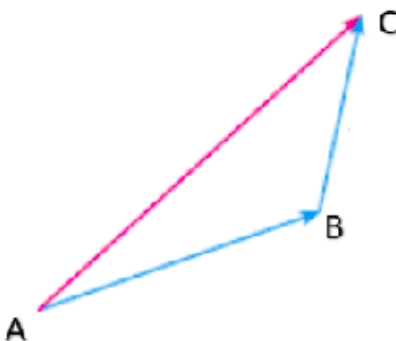


Figure 1.2.1: The triangle law

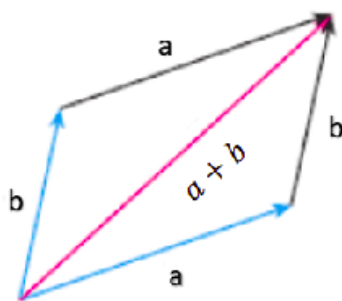


Figure 1.3.1: The parallelogram law

1.4 Components of a vector and the position vector

Consider the xy -plane or $\mathbb{R}^2$. Let P be a point in the xy -plane. Suppose the coordinates of P are (a_1, a_2) . Let O denote the origin of the xy -plane. If we drop a perpendicular from the point P to the x -axis (say, PL), then the length of OL is a_1 and the length of PL is a_2 . See Figure 1.4.1 below for an illustration. Let $\vec{a}$ be the vector $\overrightarrow{OP}$. Then we write

$$\vec{a} = \langle a_1, a_2 \rangle,$$

where a_1 and a_2 are called the **components** of the vector $\vec{a}$. We refer to a_1 as the component along the x -axis and similarly for a_2 .

Similarly, let us consider the xyz -space, and consider a point P there, whose

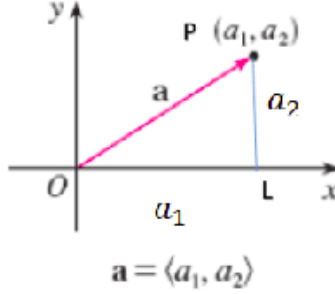


Figure 1.4.1: Components of a vector in a plane

coordinates are (a_1, a_2, a_3) . Let PL denote the perpendicular from the point P on the xy -plane. Let M denote the foot of the perpendicular from the point L on the x -axis and N denote the foot of the perpendicular from the point L on the y -axis. Then we have $PL = a_3, OM = a_1, ON = a_2$. If $\overrightarrow{OP} = \vec{a}$, then we write

$$\vec{a} = \langle a_1, a_2, a_3 \rangle,$$

where a_1, a_2, a_3 are called the **components of** the vector $\vec{a}$. See Figure 1.4.2 below for an illustration.

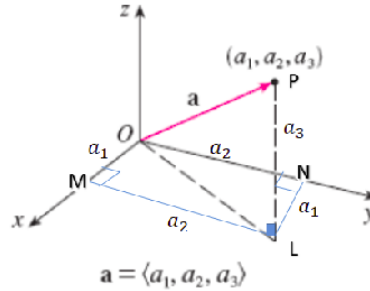


Figure 1.4.2: Components of a vector in space

For any positive integer $n > 3$, one can similarly define $a_1, \dots, a_n$ to be the **components of** an n -dimensional vector $\vec{a} = \langle a_1, \dots, a_n \rangle$.

Definition 1.4.1. If we consider a point P in the xy -plane (or in the xyz -space) and if O denotes the origin, then the vector $\overrightarrow{OP}$ is called the **position vector** of the point P .

1.5 Geometrical interpretation of addition of two vectors

Let $\vec{a}$ and $\vec{b}$ be the vectors $\langle a_1, a_2 \rangle$ and $\langle b_1, b_2 \rangle$ respectively in the xy -plane. Then we **claim** that:

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2 \rangle.$$

That is, we have to prove that the components of the vector $\vec{a} + \vec{b}$ are given by $a_1 + b_1$ and $a_2 + b_2$ respectively.

Proof of claim: We can make a parallel displacement of the vector $\vec{b}$ and bring it to a place such that the terminal point of $\vec{a}$ equals the initial point of $\vec{b}$. Then we know how to draw the vector $\vec{a} + \vec{b}$. The following figure (Figure 1.5.1) illustrates the entire procedure and also explains what are the components of the vector $\vec{a} + \vec{b}$, completing the proof of the claim.

This whole concept can be easily extended to 3 dimensions. Also, we can

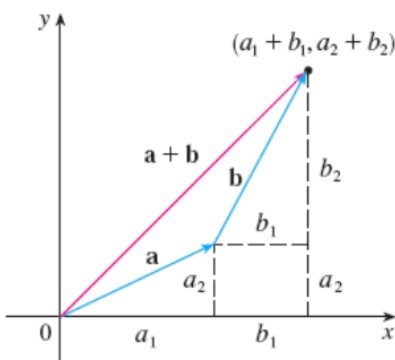


Figure 1.5.1: Components of $\vec{a} + \vec{b}$

similarly have a geometrical interpretation of the following **claim** in a similar way:

Claim: $\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2 \rangle$.

1.5.1 An application

Now suppose we consider two points P and Q in the xyz -space. Say, the coordinates of P are (x_1, y_1, z_1) and that of Q are (x_2, y_2, z_2) . Let O denote the origin of the xyz -space. Then $\vec{OP}$ is the position vector of P and $\vec{OQ}$ is the position vector of Q . If P and Q are joined, then from the definition of addition of vectors, we can see that

$$\vec{OQ} = \vec{OP} + \vec{PQ}.$$

In other words,

$$\vec{PQ} = \vec{OQ} - \vec{OP}.$$

In terms of components, we therefore have:

$$\vec{PQ} = \langle x_2, y_2, z_2 \rangle - \langle x_1, y_1, z_1 \rangle.$$

But then, using one of the above claims, we have that:

$$\vec{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

So, to conclude, if we want to express a vector $\vec{PQ}$ where P and Q are two points lying in the xyz -space, then if we know the coordinates of the points P and Q , then $\vec{PQ}$ can be written as above ($\vec{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$) in terms of components.

1.6 Magnitude or length of a vector

Let P be a point in the xy -plane, whose coordinates are (a_1, a_2) . Let O denote the origin of the xy -plane. Then $\vec{OP} = \langle a_1, a_2 \rangle$. Let L denote the foot of the perpendicular from the point P on the x -axis. Then we know that the length of OL equals a_1 and the length of PL equals a_2 . Consider the right angled triangle OPL . By Pythagoras theorem, the length of OP equals $\sqrt{a_1^2 + a_2^2}$. This length is denoted by $|\vec{OP}|$ and is called the **magnitude** of the position vector $\vec{OP}$.

Similarly, in 3 dimensions, if the coordinates of a point P are (a_1, a_2, a_3) , then the magnitude of the position vector $\vec{OP}$ is given by

$$|\vec{OP}| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

For a quick proof of this, let L denote the foot of the perpendicular from the point P on the xy -plane. Then the coordinates of L are $(a_1, a_2, 0)$ and the length of $\overrightarrow{PL}$ equals a_3 . Let A denote the foot of the perpendicular from L on the x -axis and B denote the foot of the perpendicular from L on the y -axis. Then the length of $\overrightarrow{OA}$ equals a_1 and the length of $\overrightarrow{OB}$ equals a_2 and $|\overrightarrow{OL}| = \sqrt{a_1^2 + a_2^2}$. Now, consider the right angled triangle OLP . Then $|\overrightarrow{OP}| = \sqrt{|\overrightarrow{OL}|^2 + |\overrightarrow{PL}|^2}$. That is,

$$|\overrightarrow{OP}| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

1.6.1 A remark

Till now, we have considered the length or magnitude of only position vectors (which start at the origin and end at some point). Recall from our earlier discussion that

$$\overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle,$$

where $\overrightarrow{PQ}$ was an arbitrary vector lying in the 3 dimensional space, which was obtained by joining the two points P and Q . What about the magnitude of $\overrightarrow{PQ}$?

To answer this question, recall that we can bring the vector $\overrightarrow{PQ}$ to a place such that it starts at the origin and ends at some point, using a parallel displacement.

2 An exercise

Let n be any positive integer. Let V_n denote the set of all n -dimensional vectors. Prove the following properties of vectors (by taking the vectors in terms of their components):

Properties of Vectors If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in V_n and c and d are scalars, then

1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$

2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$

3. $\mathbf{a} + \mathbf{0} = \mathbf{a}$

4. $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$

5. $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$

6. $(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$

7. $(cd)\mathbf{a} = c(d\mathbf{a})$

8. $1\mathbf{a} = \mathbf{a}$

3 Standard basis vectors

Before we begin with the topic “Standard basis vectors”, we will need the definition of a “unit vector”.

Definition 3.0.1. Any vector having magnitude 1 is called a **unit vector**. We generally denote a unit vector by a hat (or cap) over a mathematical symbol. $\square$

Remark 3.0.2. Let $\vec{a}$ be an arbitrary vector. Then $|\vec{a}|$ denotes the magnitude of $\vec{a}$. Let $\hat{a}$ denote the unit vector along $\vec{a}$. Clearly then $\vec{a} = |\vec{a}|\hat{a}$. In other words, $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$.

In the xyz -space, consider 3 points along the x -axis, y -axis, and z -axis, in such a way that their coordinates are $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ respectively. Let A, B, C denote these 3 points respectively. Let O denote the origin of the xyz -space. Then

$$\vec{OA} = \langle 1, 0, 0 \rangle$$

$$\vec{OB} = \langle 0, 1, 0 \rangle$$

$$\vec{OC} = \langle 0, 0, 1 \rangle.$$

It is easy to see that

$$|\vec{OA}| = |\vec{OB}| = |\vec{OC}| = 1.$$

So we are getting 3 unit vectors along the x, y and z axes, namely $\vec{OA}, \vec{OB}, \vec{OC}$. There is a specific way of designating these vectors. We denote $\vec{OA}, \vec{OB}, \vec{OC}$

by $\hat{i}, \hat{j}, \hat{k}$ respectively. These 3 unit vectors $\hat{i}, \hat{j}, \hat{k}$ are called the **Standard basis vectors**. Figure 3.0.1 gives a portrait of these standard basis vectors. **Question:** Why are $\hat{i}, \hat{j}, \hat{k}$ called the standard basis vectors?

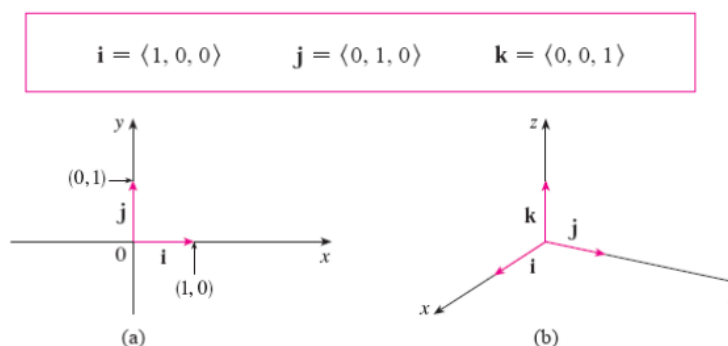


Figure 3.0.1: The standard basis vectors

Answer: Let us look at an arbitrary position vector $\overrightarrow{OP} = \langle a_1, a_2, a_3 \rangle$ in $\mathbb{R}^3$. Then observe that

$$\begin{aligned}\overrightarrow{OP} &= \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}.\end{aligned}$$

So any position vector in $\mathbb{R}^3$ can be expressed in terms of (or as a “linear combination of ”) the 3 unit vectors $\hat{i}, \hat{j}, \hat{k}$ [In fact, a position vector in $\mathbb{R}^3$ can be expressed as a “linear combination” of some other “basis vectors” as well. But $\hat{i}, \hat{j}, \hat{k}$ are called the “standard basis vectors” because they are the unit vectors along the x , y , and z -axes respectively.]. The word “standard” is used because these vectors are along the axes. Figure 3.0.2 explains the above fact(s).

4 Resolving a vector in 2-dim into components

Suppose $\vec{a}$ is an arbitrary vector. Suppose we want to resolve the component of this vector $\vec{a}$ along a certain direction. For sake of simplicity, we may assume that particular direction to be the x -axis.

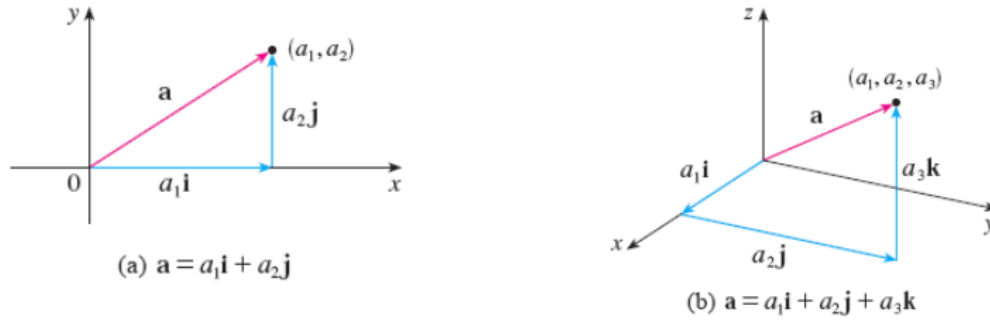


Figure 3.0.2: Why the name “standard basis vectors”?

Suppose the vector $\vec{a}$ makes an angle θ with the x -axis. If we consider the initial point of the vector $\vec{a}$ as the origin, then we can easily draw the y -axis in a direction perpendicular to the x -axis. Let O denote the origin, which is also the initial point of the vector $\vec{a}$. Let P denote the terminal point of the vector $\vec{a}$. Then $\overrightarrow{OP} = \vec{a}$. The magnitude of the vector $\vec{a}$ is given by $|\vec{a}|$. Let L denote the foot of the perpendicular from the point P on the x -axis. Then

$$\text{Length of } \overrightarrow{LP} = |\vec{a}| \sin \theta$$

and

$$\text{Length of } \overrightarrow{OL} = |\vec{a}| \cos \theta.$$

Then we know that

$$\overrightarrow{OL} = |\vec{a}| \cos \theta \hat{i}$$

and

$$\overrightarrow{LP} = |\vec{a}| \sin \theta \hat{j}.$$

So eventually,

$$\begin{aligned} \vec{a} &= \overrightarrow{OP} = \overrightarrow{OL} + \overrightarrow{LP} \\ &= |\vec{a}| \cos \theta \hat{i} + |\vec{a}| \sin \theta \hat{j}. \end{aligned}$$

In this way, we can resolve the components of a vector. If the coordinates of the point P are (a_1, a_2) , then $a_1 = |\vec{a}| \cos \theta$ and $a_2 = |\vec{a}| \sin \theta$.

4.1 An example

A 100-lb weight hangs from two wires as shown in Figure 4.1.1. Find the tensions (forces) T_1 and T_2 in both wires and the magnitudes of the tensions.

The resultant $\vec{T}_1 + \vec{T}_2$ of the tensions counterbalances the weight $\vec{W}$. Since

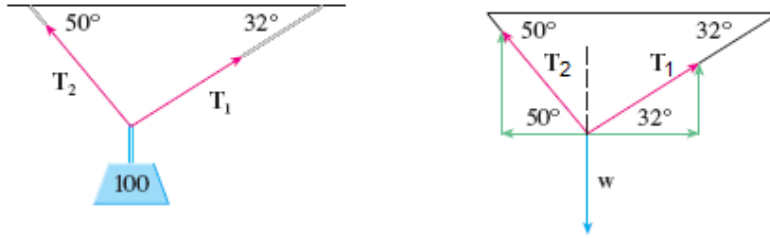


Figure 4.1.1: Example 4.1

the sum of $\vec{T}_1$ and $\vec{T}_2$ balances the weight $\vec{W}$, the resultant vector $\vec{T}_1 + \vec{T}_2$ must be equal in magnitude to $\vec{W}$ and have direction opposite to it. In fact, one can easily observe that the sum vector $\vec{T}_1 + \vec{T}_2$ will be in the vertically upward direction. So from Figure 4.1.1, we must have that

$$\vec{T}_1 + \vec{T}_2 = -\vec{W} = 100\hat{j}.$$

Let us consider the junction of T_1 and T_2 as the origin O . Then we have:

$$\vec{T}_1 = |\vec{T}_1| \cos 32^\circ \hat{i} + |\vec{T}_1| \sin 32^\circ \hat{j} \quad (i)$$

$$\vec{T}_2 = -|\vec{T}_2| \cos 50^\circ \hat{i} + |\vec{T}_2| \sin 50^\circ \hat{j} \quad (ii)$$

and $\vec{W} = -100\hat{j}$.

Now $\vec{W} = \vec{T}_1 + \vec{T}_2$ gives

$$|\vec{T}_1| \cos 32^\circ - |\vec{T}_2| \cos 50^\circ = 0 \quad (a)$$

and

$$|\vec{T}_1| \sin 32^\circ + |\vec{T}_2| \sin 50^\circ = 100. \quad (b)$$

From (a), we get

$$|\vec{T}_1| = |\vec{T}_2| \frac{\cos 50}{\cos 32}. \quad (*)$$

Substituting $|\vec{T}_1|$ in (b), we can get $|\vec{T}_2|$. Then from (*), we can get $|\vec{T}_1|$. Now putting the values of $|\vec{T}_1|$ and $|\vec{T}_2|$ in (i) and (ii), we can get the tensions $\vec{T}_1$ and $\vec{T}_2$.

5 The dot product of two vectors

Definition 5.0.1. Let $\vec{a} = \langle a_1, a_2, a_3 \rangle = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. Then the **dot product** of the vectors $\vec{a}$ and $\vec{b}$ is denoted by $\vec{a} \cdot \vec{b}$ and is defined as

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

□

Remark 5.0.2. From the above expression of the dot product $\vec{a} \cdot \vec{b}$, we can easily see that $\vec{a} \cdot \vec{b}$ is a scalar [A scalar is a physical quantity having only magnitude.]. That's why the dot product is also called the "scalar product".

Remark 5.0.3. Note that $\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2 = |\vec{a}|^2$.

Let V_3 denote the set of all 3-dimensional vectors. Let $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ be vectors in V_3 and c be a scalar. Then the following properties (as mentioned in Figure 5.0.1) of the dot product are obvious:

- | | |
|---|---|
| 1. $\mathbf{a} \cdot \mathbf{a} = \mathbf{a} ^2$ | 2. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ |
| 3. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ | 4. $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$ |
| 5. $\mathbf{0} \cdot \mathbf{a} = 0$ | |

Figure 5.0.1: Properties of the dot product

5.1 Geometrical interpretation of the dot product

Consider two vectors $\vec{a}$ and $\vec{b}$ having the same initial point. [Even if, $\vec{a}$ and $\vec{b}$ may not have the same initial point to begin with, we can make them so by using parallel displacement.] Let θ be the angle between $\vec{a}$ and $\vec{b}$. Figure 5.1.1 illustrates the scenario. Then we have:

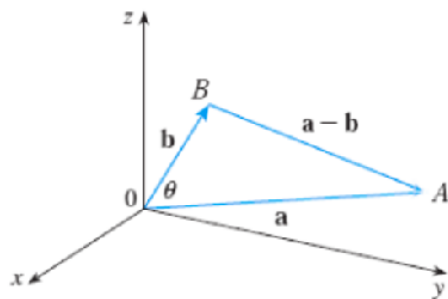


Figure 5.1.1: Geometrical interpretation of the dot product

$$\begin{aligned}
 |\vec{a} - \vec{b}|^2 &= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) \\
 &= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\
 &= |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2
 \end{aligned} \tag{1}$$

where the last equality is due to the fact that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$.

You must have also studied in geometry/trigonometry that using the properties of a triangle, we can write:

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos\theta. \tag{2}$$

From equations (1) and (2), we get:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta.$$

This is the geometrical interpretation of the dot product.

Remark 5.1.1. Clearly, the quantity $|\vec{a}||\vec{b}|\cos\theta$ is always ≥ 0 . It follows from the above geometrical interpretation that:

- (i) If θ is acute, then $\vec{a} \cdot \vec{b} > 0$.
- (ii) If $\theta = \frac{\pi}{2}$, then $\vec{a} \cdot \vec{b} = 0$.
- (iii) If θ is obtuse, then $\vec{a} \cdot \vec{b} < 0$.

Remark 5.1.2. If $\vec{a} \cdot \vec{b} = 0$, then $\cos \theta = 0$, which in turn implies that $\theta = (2n+1)\frac{\pi}{2}$ for some integer n . Then the vectors $\vec{a}$ and $\vec{b}$ are “orthogonal”. Hence $\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a}$ and $\vec{b}$ are orthogonal.

Remark 5.1.3. The angle θ between $\vec{a}$ and $\vec{b}$ can be found from the geometrical interpretation as:

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \right).$$

6 Direction angles and direction cosines

Recall that when we consider a vector $\vec{a} = \langle a_1, a_2 \rangle$ in 2-dimensions and we try to resolve its components along the x -axis and the y -axis, then

$$a_1 = |\vec{a}| \cos \theta$$

and

$$a_2 = |\vec{a}| \sin \theta,$$

where θ is the angle made by the vector $\vec{a}$ with the x -axis.

Question: Can we extend these ideas to 3-dimensions?

Answer: Yes. This brings in the concept of direction angles and direction cosines.

We will now consider an arbitrary vector in 3-dimensions, whose initial point is the origin O and the terminal point is P . Even though, the initial point of the vector considered is not at the origin to begin with, we can make it so by using parallel displacement.

Now we are going to consider 3 orthogonal axes at the origin O . Let the point P have coordinates (a_1, a_2, a_3) . Let $\vec{OP}$ make an angle α with the x -axis, β with the y -axis, and γ with the z -axis. Figure 6.0.1 explains the situation here. Let us consider the standard basis vectors $\hat{i}, \hat{j}, \hat{k}$ along the x, y and z -axes respectively. Let $\vec{OP} = \vec{a}$. Then

$$\vec{OP} = \vec{a} = \langle a_1, a_2, a_3 \rangle = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}.$$

Observe that if we consider the dot product of $\hat{i}$ with $\vec{a}$, then we get:

$$\hat{i} \cdot \vec{a} = a_1,$$

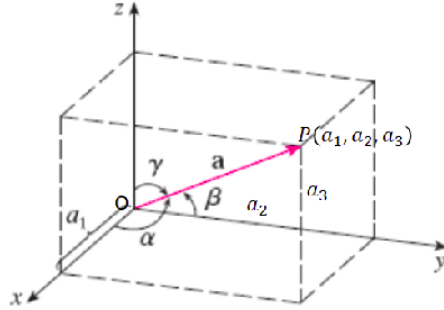


Figure 6.0.1: Direction angles

which is the same as saying that

$$|\hat{i}| |\vec{a}| \cos \alpha = a_1.$$

$$\Rightarrow 1 \cdot |\vec{a}| \cos \alpha = a_1$$

$$\Rightarrow \cos \alpha = \frac{a_1}{|\vec{a}|}.$$

Similarly, we can get:

$$\cos \beta = \frac{a_2}{|\vec{a}|} \text{ and } \cos \gamma = \frac{a_3}{|\vec{a}|}.$$

These 3 angles α, β and γ are called the **direction angles**. The cosines of α, β and γ are called the **direction cosines**.

Remark 6.0.1. Observe that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{a_1^2 + a_2^2 + a_3^2}{|\vec{a}|^2} = \frac{|\vec{a}|^2}{|\vec{a}|^2} = 1.$$

Recall that

$$\cos \alpha = \frac{a_1}{|\vec{a}|}, \cos \beta = \frac{a_2}{|\vec{a}|}, \cos \gamma = \frac{a_3}{|\vec{a}|}.$$

Now

$$\begin{aligned} \vec{a} &= a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \\ &= |\vec{a}| (\cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}). \end{aligned}$$

So the vector $\vec{a}$ can be expressed in this way as well!
 From the above, we also get that

$$\frac{\vec{a}}{|\vec{a}|} = \langle \cos \alpha, \cos \beta, \cos \gamma \rangle .$$

Observe that $\frac{\vec{a}}{|\vec{a}|}$ is a unit vector. So $(\cos \alpha, \cos \beta, \cos \gamma)$ are nothing but the components of the unit vector along $\vec{a}$.

7 Projections

There are 2 types of projections that we will consider in this section. One is called the “vector projection”, and the other one is called the “scalar projection”.

Definition 7.0.1. Suppose we consider 2 vectors $\vec{a}$ and $\vec{b}$ having the same initial point. Suppose θ is the angle between them. Let $\vec{a} = \overrightarrow{PQ}$ and $\vec{b} = \overrightarrow{PR}$. Suppose we draw a perpendicular from the point R on PQ . Let RS denote this perpendicular. Then the length PS is the “shadow” of the vector $\vec{b}$ over the vector $\vec{a}$. The vector $\overrightarrow{PS}$ is called the **vector projection** of the vector $\vec{b}$ over the vector $\vec{a}$. The signed magnitude $|\vec{b}| \cos \theta$ of $\overrightarrow{PS}$ is called the **scalar projection** of the vector $\vec{b}$ over the vector $\vec{a}$. The vector $\overrightarrow{PS}$ is denoted by $proj_{\vec{a}} \vec{b}$ and the signed magnitude $|\vec{b}| \cos \theta$ of $\overrightarrow{PS}$ is denoted by $comp_{\vec{a}} \vec{b}$. $\square$

Figure 7.0.1 illustrates the above definition. Then

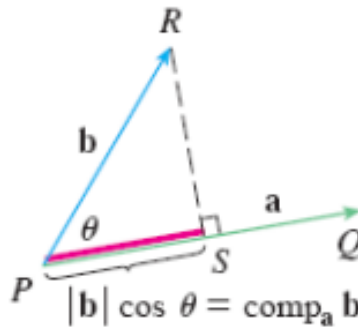


Figure 7.0.1: vector and scalar projections

$$|\vec{b}| \cos \theta = \frac{|\vec{a}||\vec{b}| \cos \theta}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \text{comp}_{\vec{a}} \vec{b}.$$

Let us now consider the unit vector $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$ in the direction of the vector $\vec{a}$. Clearly, $\hat{a}$ is a unit vector along the direction of $\vec{a}$. Then

$$\begin{aligned} \vec{PS} &= (\text{signed magnitude of } \vec{PS})(\text{a unit vector along } \vec{PS}) \\ &= (\text{signed magnitude of } \vec{PS})\hat{a} \\ &= (\text{signed magnitude of } \vec{PS})\frac{\vec{a}}{|\vec{a}|} \\ &= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \right) \frac{\vec{a}}{|\vec{a}|} = \frac{(\vec{a} \cdot \vec{b})\vec{a}}{|\vec{a}|^2}. \end{aligned}$$

Observe that $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2}$ is a scalar. So $\text{proj}_{\vec{a}} \vec{b}$ is a vector, but $\text{comp}_{\vec{a}} \vec{b}$ is a scalar.

Remark 7.0.2. In Figure 7.0.1, the angle θ was acute. What happens if θ is an obtuse angle? Figure 7.0.2 explains the situation when θ is an obtuse angle. Always remember that we have to take the perpendicular from the terminal point

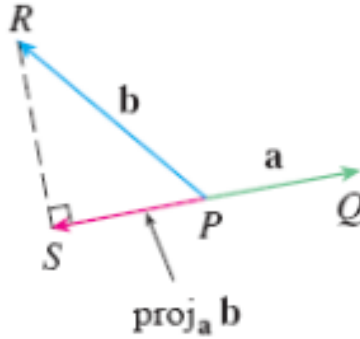


Figure 7.0.2: When the angle between $\vec{a}$ and $\vec{b}$ is obtuse

of the vector $\vec{b}$. In the case of an obtuse angle, we need to extend the vector $\vec{PQ}$ in the opposite direction. And we are going to consider PS (see Figure 7.0.2) for our vector and scalar projections.

7.1 An application

Suppose we are trying to drag an object from the point P along the horizontal direction, with a force (say $\vec{F}$) in the direction (as shown in Figure 7.1.1) and the object is moved from the point P to the point Q . Then $\overrightarrow{PQ} =$

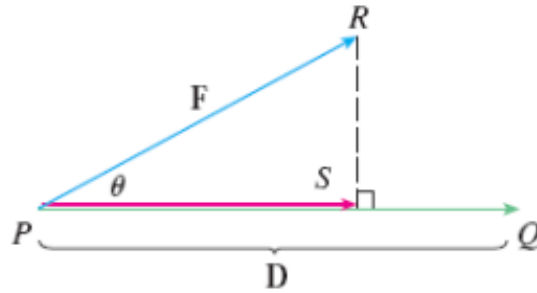


Figure 7.1.1: An application

the displacement vector. The dragging from P to Q (along the horizontal direction) was accomplished by the component of $\vec{F}$ along the horizontal direction. This component is $|\vec{F}| \cos \theta$. We know that the work done W by the force $\vec{F}$ in moving the object from P to Q is given by

$$\begin{aligned} W &= (|\vec{F}| \cos \theta) |\overrightarrow{PQ}| \\ &= |\vec{F}| \cos \theta |\vec{D}| = \vec{F} \cdot \vec{D}, \end{aligned}$$

where $\vec{D} = \overrightarrow{PQ}$.

Although this is not a direct application of the projections, but one can express the above (the work done W) easily in terms of projections.

—————-END—————